

INDIAN SCHOOL OF BUSINESS

LLMs and Generative AI (LGAI)

Group Assignment

Honor Code Scheme: **2N-b**

Weightage: **40%**

Deadline: **07th June 2026, 11:55PM**

Deliverables:

- A report (**A pdf file**) with answers and outputs for **PART A, B, C and D.**
- **5 Python (.ipynb)** code files. (**Retain the outputs in the notebooks and** give proper comments/explanations/interpretations for the code.
- Assignment Submission Form (**should be submitted separately**)

General Instructions:

1. This is a Group Assignment. The **maximum marks** are **40**.
2. The file should be named “Group-nn” where nn is your group number.
3. Make sure that only one person from the group submits the assignment.
4. Type your group member’s name on the first page of the Assignment. Marks will not be awarded if the name(s) are missing.
5. Please do not submit .zip files, the submission will not be considered.
6. Code files rendered/exported as pdfs will strictly not be considered for evaluation.
7. Any **late submission will attract a penalty** as mentioned in the course outline
8. The honor code for this submission is **2N-b**. Please look through the honor code restrictions carefully before attempting the assignment as there will be strong consequences for breaking them.
9. **Email submissions are not allowed. All the submissions must be made on the LMS.**
10. Upload your submissions to the ‘**LGAI Group Assignment**’ folder on LMS.
11. Peer evaluation scores will be applied to the Group Assignment scores.
12. **Please adhere to the given instructions, otherwise, your submission will not be accepted, or a severe penalty will be applied.**

Assignment Questions:

Part A:

List down four scenarios where each of these could be useful. **[8 marks]**

1. Image Classification
2. Image similarity computation
3. Object Detection
4. Image Segmentation
5. Visual Question Answering
6. Action Recognition
7. Multi-object tracking
8. Image and video generation

A scenario should contain business domain, input data, expected output, some company which has deployed this scenario. Also list down a paper/blog reference if you can find one.

An example of Object Detection:

- Domain: Transportation
- Input data: Railway traffic dataset
- Expected output: Objects of these types – Spanner, helmet, pedestrian, railway left class, railway right class, bullet train, railway straight.
- Company which has deployed: China Railway Bureau
- Paper/blog/case studies reference (if any):
https://www.researchgate.net/publication/343561230_Autonomous_Railway_Traffic_Object_Detection_Using_Feature-Enhanced_Single-Shot_Detector

Part B:

You are given a **dataset** with 15 categories. There are a total of 805 images.

1. For each class, treat images from 0001 to 0040 as train and the remaining as test.
2. Using pretrained resnet18, densenet121, vgg19 models on this dataset, train a 15-class classification model and report per-class classification accuracy in terms of precision and recall. **Submit q1.ipynb. [8 marks]**
3. Finetune the vgg19 model on this dataset, train a 15-class classification model and report per-class classification accuracy in terms of precision and recall. **Submit q2.ipynb. [4 marks]**
4. Also, report zero-shot per-class classification accuracy in terms of precision and recall with CLIP-ViT-B/16 model. **Submit q3.ipynb. [4 marks]**

Part C:

You are given a dataset with 84 **pathology** images. There are also associated mask images with nuclei marked on them. Use 60 for train and remaining for test.

Finetune UNet model with ResNet-18 backbone for segmentation on 24 test images, and report precision and recall for the nucleus class. **Submit q4.ipynb. [8 marks]**

Part D:

You are given a dataset (**dataset2.zip**) with 128 object images. You are also provided associated object annotations. Use first 100 for train and remaining for test.

The dataset contains 4 kinds of objects: Water Bottle, Milk Bottle, Tetra Pack and Can.

Finetune a pretrained Faster RCNN model for object detection on train data. On 28 test images, report precision and recall for each (except background) with an IoU threshold of 0.5. **Submit q5.ipynb. [8 marks]**

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